Computers and Geosciences 113 (2018) 33–42

Contents lists available at ScienceDirect

Computers and Geosciences

journal homepage: www.elsevier.com/locate/cageo

Research paper

Modelling soil-water dynamics in the rootzone of structured and water-repellent soils

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ARTICLE INFO

Keywords:
Critical zone
Soil-water dynamics
APSIM modelling
Preferential flow
Soil-water repellency
Evapotranspiration

ABSTRACT

In modelling the hydrology of Earth's critical zone, there are two major challenges. The first is to understand and model the processes of infiltration, runoff, redistribution and root-water uptake in structured soils that exhibit preferential flows through macropore networks. The other challenge is to parametrise and model the impact of ephemeral hydrophobicity of water-repellent soils. Here we have developed a soil-water model, which is based on physical principles, yet possesses simple functionality to enable easier parameterisation, so as to predict soil-water dynamics in structured soils displaying time-varying degrees of hydrophobicity. Our model, WEIRDO (**W**ater **E**vapotranspiration **I**nfiltration **R**edistribution **D**rainage run**O**ff), has been developed in the APSIM Next Generation platform (Agricultural Production Systems sIMulation). The model operates on an hourly time-step. The repository for this open-source code is https://github.com/APSIMInitiative/ApsimX. We have carried out sensitivity tests to show how WEIRDO predicts infiltration, drainage, redistribution, transpiration and soil-water evaporation for three distinctly different soil textures displaying differing hydraulic properties. These three soils were drawn from the UNSODA (Unsaturated SOil hydraulic Database) soils database of the United States Department of Agriculture (USDA). We show how preferential flow process and hydrophobicity determine the spatio-temporal pattern of soil-water dynamics. Finally, we have validated WEIRDO by comparing its predictions against three years of soil-water content measurements made under an irrigated alfalfa (*Medicago sativa* L.) trial. The results provide validation of the model's ability to simulate soil-water dynamics in structured soils.

1. Introduction

Earth's critical zone is the "heterogeneous, near surface environment in which complex interactions involving rock, soil, water, air, and living organisms regulate the natural habitat and determine the availability of life-sustaining resources" (National Research Council, 2001). This critical zone is a realm of geoscientific interest because the teeming hydrological processes in this near-surface environment control the quantity and quality of our water resources, as well as the productivity of the vegetation systems, both natural and managed, growing on fractured rocks, and in soils. Furthermore, management of agricultural lands often involves the use of irrigation water, and so understanding and predicting soil-water dynamics in the structured soils of the vadose zone is critical for food production, and also for protecting receiving water bodies from nutrient leaching and runoff losses.

There are two challenges in predicting the soil-water dynamics in the rootzone of structured soils: the preferential transport processes through the macropores (Jarvis et al., 2016); and the widespread and complex temporal variation in the occurrence and severity of hydrophobicity, or soil-water repellency (Müller et al., 2014).

Models, of many types, have been developed, over many years, to enable prediction of water and solute movement through structured soils, and also for transport processes in water-repellent soils. However, few models have considered preferential flow processes in water-repellent soils. Here we describe a model that incorporates both processes simultaneously, and which has been coded in the APSIM (Agricultural

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https://doi.org/10.1016/j.cageo.2018.01.014
Received 28 July 2017; Received in revised form 17 January 2018; Accepted 21 January 2018
Available online 4 February 2018
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Table 1
Schematic representation of the soil-array structure by pore cohort (*c*) and profile layer (*l*).

		Pore cohort (c)									
		c = 0	c = 1	c = 2	c = 3	c = 4	c = 5	c = 6	c = 7	c = 8	c = 9
Profile layer (l)	l = 0	(0,0)	(0,1)	(0,2)	(0,3)	(0,4)	(0,5)	(0,6)	(0,7)	(0,8)	(0,9)
	l = 1	(1,0)	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)	(1,7)	(1,8)	(1,9)
	l = 2	(2,0)	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)	(2,7)	(2,8)	(2,9)
	l = 3	(3,0)	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)	(3,7)	(3,8)	(3,9)

Production Systems sIMulator) Next Generation platform (Holzworth et al., 2015). Our model is termed WEIRDO (Water Evapotranspiration Infiltration Redistribution Drainage runOff). The repository for this open-source code is https://github.com/APSIMInitiative/ApsimX. Holzworth et al. (2015) note that APSIM Next Gen has a quicker 'time to release' for new models, along with better, more up-to-date documentation and better testing. WIERDO is one of these new models.

1.1. Preferential flow

Long-ranging, preferential flows through the macropores of structured soils have generally been described either by using functional, or mechanistic models. Addiscott (1977) developed a functional model by considering the soil as a stack of slabs, each with two domains: a matrix phase and fast-flowing macropore phase. The macropore domain connected the layers, such that water flowed vertically and/or was absorbed 'sideways' into the matrix. Mechanistic schema have used analytical descriptions to describe the physics of the Richards' flows in dual-domain, and multi-domain models of transport and exchange processes (van Genuchten and Wierenga, 1976; Gerke and Van Genuchten, 1993; Jarvis et al., 2016).

To balance between the restrictive simplicity of functional models and the onerous parametrisation requirements of mechanistic models, we have developed WEIRDO. It is a functional model which enables easier implementation and parameterisation. But it has detailed representation of the soil's porosity and draws its parameters from the soil-water characteristic and hydraulic conductivity function, thereby ensuring a rational physical basis.

1.2. Hydrophobicity

Mechanistic models of the impact of soil-water repellency on soil-water processes have been developed using detailed understanding of the macroscopic dynamics of wetting and drying, and the time-dependency of the contact-angle of wetting in hydrophobic soils (Deurer and Bachmann, 2007). Here in our model for hydrophobicity, we take all of these physical characteristics and we express them in a functional model which enables easy characterisation from the measurements that are normally made to assess water repellency (Wijewardana et al., 2015). Water-repellency was found to widespread across the pastoral soils of New Zealand (Deurer et al., 2011). Our WEIRDO model computes any runoff at the surface by creating a 'surface-pond' of water. This depth of water is then excluded from further infiltration calculations. We have not presented here any simulations dealing with such runoff, which occurs when the applied flux of rain, or irrigation, exceeds the ability of the surface soil's porous system to accept the incident rate. In future, we will connect WEIRDO with a spatially-explicit version of APSIM where will route runoff across the surface according to the depth of the surface pond, the local slope, the surface detention-capacity, and flow resistance.

2. Model structure

Our model, WEIRDO has been developed in the APSIM Next Generation platform to enable easy integration with crop, management and climate modules, and to provide an interface for the easy set-up, execution and visualisation of simulations. It is a layered soil-water balance model designed to capture bypass flow phenomena and the impacts of hydrophobicity, which are often observed but not captured by many soil-water models. Details of the parameters and their calculation, are given in Tables 1 and 2.

2.1. Pore characteristics

The total porosity of each layer is partitioned into cohorts of similar sized pores more cohorts in the near-saturated region (Fig. 1). The movement of water through each cohort of pores is modelled separately. It operates at an hourly time step in most cases but reduces to a 6-min time step when high rates of irrigation or rainfall occur.

2.2. Software structure

The central notion of WEIRDO is that a soil layer's porosity can be grouped into a series of cohorts of similar-sized pores and that the overall behaviour is the result of water movement through each of these pore cohorts. At the top-level, the soil is modelled by WEIRDO.cs which links all processes and collates the pore data-structures. The soil is represented by a two-dimensional array where the first dimension is layers (*l*) in the soil and the second dimension is the pore cohorts (*c*) within the layer (Table 1). Each array member (*l*,*c*) is represented by a generic pore class (Pore.cs) which contains a set of properties defining the parameters of the pore cohort and keeps track of its state.

Each of these properties and their methods of calculation are detailed in Table 2. The only property that is set is the equivalent depth of the water volume stored in each pore cohort (D_W). All other properties are either parameters, set when the model is initialised, or are a function of D_W and/or other parameters. Whenever a process is calculated, WEIRDO.cs will step through each pore class in each layer to determine the changes in the water-in and water-out. The number of pore compartments is fixed in the code at 10 per layer. The soil is sliced into 50 mm layers, and the number of layers (*Nl*) and subsequent profile depth is arbitrary.

2.3. Pore set-up

The model requires the water contents at saturation ($\theta_{S,}$ 0 mm pressure head, ψ), drained upper limit ($\theta_{D,}$ $-1.0e^3$ mm pressure head, ψ), lower limit (θ_L, $-1.5e^5$ mm pressure head, ψ) and air dry (θ_A, $-6.0e^7$ mm pressure head, ψ). A bubbling, or air-entry water potential head, (h_{bub}), is also required, and θ_S is assumed at this potential to give a 5th point to fit (Fig. 1). The scheme assumes boundaries between pore cohorts at pore diameters (d) of 3000, 1194, 475, 189, 75, 30, 8.6, 2.47, 0.707, 0.202 and 0.0005 µm (Fig. 1). So the corresponding water potentials, when all of the pores below these boundaries are water filled (h in mmH_2O), are given by −30000/d (Loll and Moldrup, 2000).

2.4. The hydraulic-flow model

The hydraulic flows in WEIRDO take into account: the vertical gravitational Poiseuille flow governed by the pore's radius which determines its hydraulic conductivity, *k*; the capillary attraction of water due to the pore's sorptivity, *S*; and the impact of hydrophobicity, which is applied only through a reduction factor on the pore's hydrophilic

Table 2
The list of properties in the generic pore class. Each pore cohort in each layer contains all of these properties and parameters.

Property	Unit	Description	Value
Descriptors			
L	–	Soil layer identifier	0 – number of layers *-1*
C	–	Pore cohort identifier	0–9
T	mm	Thickness of the current soil layer	50
H	mm	Height of layer above a datum	From soil configuration
Geometry			
d_U	μm	Diameter of largest pores in a cohort	From soil configuration
d_L	μm	Diameter of smallest pores in a cohort	From soil configuration
R	μm	Radius of a cohort's mean-sized pore	$(d_U + d_L)/4$
A	μm^2	Area of mean pore column	$\pi * r^2$
N_P	m^{-2}	Number of pore columns in the soil cross-section	$V_P/(A/1 \times 10^{12})$
Porosity			
h_U	mm	Water potential when this cohort is full	$-30000/d_U$
h_L	mm	Water potential when this cohort is just empty	$-30000/d_L$
θ_U	$m^3\ m^{-3}$	Porosity at the top boundary of this cohort	From h_U and moisture release curve (Fig. 1)
θ_L	$m^3\ m^{-3}$	Porosity at the lower boundary of this cohort	From h_L and moisture release curve (Fig, 1)
V_P	$m^3\ m^{-3}$	Total water volume of a pore cohort	$\theta_U - \theta_L$
V_A	$m^3\ m^{-3}$	Air volume of a pore cohort	V_P-V_W
V_W	$m^3\ m^{-3}$	Water volume of a pore cohort	D_W/T
D_P	mm	Equivalent depth of pore volume	V_P*T
D_A	mm	Equivalent depth of air volume	V_A*T
D_W	mm	Equivalent depth of water volume	Set by model processing
D_D	mm	Depth of water volume	D_W*F_G
Hydraulics			
X_F	–	Coefficient	From soil configuration
C_F	–	Coefficient	From soil configuration
k_{ppoi}	$mm^3\ s^{-1}$	Conductivity of a single pore column	$C_F*r^{X_F}$
k_{poi}	$mm\ h^{-1}$	Poiseuille flow of a pore cohort	$(k_{ppoi} * N_P)/1000*3600$
S	$mm\ \sqrt{s}^{-1}$	Sorptivity of a pore cohort	$\sqrt{\frac{7.4/r*1000*k_{poi}*\ V_A}{0.5}}$
k_{sor}	$mm\ h^{-1}$	Sorption of water into a pore cohort	$0.5*S*\ \sqrt{t}^{-1}$
R_F	–	Relative repellence effect on water movement into a pore cohort operating through the sorptivity only	
k_{in}	$mm\ h^{-1}$	Flow into a hydrophobic pore cohort	$k_{poi} + (k_{sor} * R_F)$
k_{out}	$mm\ h^{-1}$	Flow out a of pore cohort	$k_{poi} * F_G$
F_G	–	Relative gravitational effect on water movement out of a pore cohort	$\begin{array}{ll} 0 & P_G \geq \psi_U \\ 1 & P_G < \psi_U \end{array}$
k_{dif}	$mm\ h^{-1}$	Potential redistribution out of a pore cohort	$k_{poi}*V_W/V_P*(1 - F_G)$
P_G	mm	Gravitational potential for a layer	$(H + T)/-0.1022$
Root Water Extraction			
WE_{pot}	$mm\ h^{-1}$	Potential water extraction from a pore cohort	$D_W * P_U$
P_U		Proportion of water that may be extracted	$(k_{poi}/Dd_\mu) * 10$
Dd_μ	mm	Mean diffusion distance from soil to root	$\sqrt{1/rld}*0.5$
Rld	mm mm^{-3}	Crop root length density	From the crop model

sorptivity since it is essentially a capillary wetting phenomenon (Clothier et al., 2000).

2.4.1. Poiseuille flow and capillarity

The conductivity model for gravity flow relates specifically to Poiseuille flow through a cylindrical pore, as measured when infiltration is at steady-state through a soil layer, and when sorption and repellency processes are negligible. The Poiseuille conductivity of a soil layer ($k_{poi(l)}$) depends on its pore-size distribution and this is handled in WEIRDO using the model described by Arya et al. (1999). Each cohort of pores is modelled as a collection of parallel, continuous tubes, each with an individual Poiseuille flow ($k_{ppoi(l,c)}$) that is dependent on the mean radius of the pores in that cohort (Table 2). The overall conductivity of the cohort is also dependent of the number of pores it contains (N_P, Table 2) and the conductivity of the layer is dependent on which pore cohorts currently contain water, which depends on the pressure potential head, ψ.

The model requires a further two dimensionless parameters for each layer of soil, namely X_F and C_F which are empirical and determined via a fitting exercise. For the development of WEIRDO we fitted a 5-point Hermite spline to soil-water characteristic, $\theta(\psi)$ (Huth et al., 2012), and the conductivity curve, $K(\psi)$ is of the form described by Arya et al. (1999). We tested this on 20 soils from the UNSODA data-base (https://data.nal.usda.gov/dataset/unsoda-20-unsaturated-soil-hydraulic-database-database-and-program-indirect-methods) to determine the adequacy of the fits and the range in the X_F and C_F parameters. Three fitted splines are shown in Fig. 2 for three quite different soils from UNSODA data-base: Soil 4650 (red), a sand, with a high macroporosity of 0.28 $m^3\ m^{-3}$ and saturated conductivity k_{sat} of 100 mm h^{-1}; Soil 4671 (green), a silt, with a macroporosity of 0.05 $m^3\ m^{-3}$ and k_{sat} of 5 mm h^{-1}; and Soil 4602 (blue), a loam, with a macroporosity of 0.05 $m^3\ m^{-3}$ and a k_{sat} of 0.03 mm h^{-1}. We use these three 'soils' in a series of sensitivity tests with WEIRDO.

The hydraulic properties of the sand, silt and loam are quite different, and the fitted X_F and C_F values (Fig. 2) reflect these differences. In Fig. 3, as a heuristic exercise we show how different X_F and C_F values influence the hydraulic conductivity function $k(\psi)$. The parameter X_F represents the slope of the hydraulic conductivity function, $k(\psi)$ (Fig. 3), whereas C_F controls the conductivity of the smallest pore cohort of pore of diameter less than 0.202 μm.

The saturated hydraulic conductivity of a layer, $k_{sat(l)}$, is a critical soil property, and it is often known. A particular value of $k_{sat(l)}$ can be achieved by adjusting $C_{F(l)}$, $X_{F(l)}$, $h_{bub(l)}$, $\theta_{S(l)}$, $\theta_{D(l)}$, and $\theta_{L(l)}$. However, it is important to ensure sensible $k(\psi)_{(l)}$ are fitted, because these influence absorption, diffusion and crop water uptake processes. It is ideal to have measurements of $h_{bub(l)}$, $\theta_{S(l)}$, $\theta_{D(l)}$, $\theta_{L(l)}$ $k_{sat(l)}$ and $k(\psi)_{(l)}$ at some unsaturated values to enable the fitting of sensible values for $C_{F(l)}$ and $X_{F(l)}$. Presently, this fitting procedure is done external to APSIM using a set of Python scripts (www.python.org), but in future this initial parameterisation will be built into the APSIM user interface.

As well as Poiseuille flow due to gravity, water moves through soil due to capillary forces from the small pores of the soil's matrix. The sorptivity, S, parameterises the square-root-of-time cumulative infiltration, I, into dry soil due to capillary forces (Philip, 1987): I=$St^{1/2}$. The rate of capillary sorption, i, is ½ $St^{-1/2}$, or here just ½ S because we use hourly time steps. The sorption of water into a pore cohort, $k_{sor(l,c)}$ is calculated as a function of the pore's radius $r_{(l,c)}$, $k_{poi(l,c)}$, which is inversely related to the ratio of the surface area to volume (Table 2), and the air-filled volume of the pore, $V_{A(l,c)}$ (White and Sully, 1987). Thus, capillary sorption into a pore will be greatest when the pore is air filled, and zero when the pore is water filled.

2.4.2. Hydrophobicity

The oft-observed phenomenon of soil-water repellency, or hydrophobicity, occurs when water does not enter the matrix of the soil by capillarity at a rate that would be expected from the soil's sorptivity, S, as

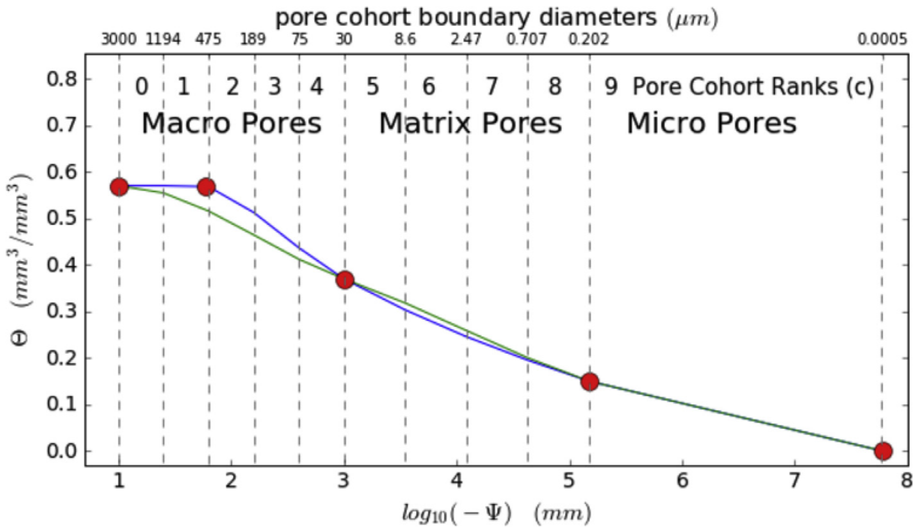

Fig. 1. An example soil-water release curve showing how the porosity of a soil is divided into pore cohorts. Symbols are required input parameters for the volumetric water content (θ) at saturation, air-entry potential, −1000 mm potential head, −150,000 mm potential head, and air dry. The green line is a fitted Hermite spline as described by Huth et al. (2012) and the blue line is a modified version which includes a fifth point at the bubbling, or air-entry potential. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

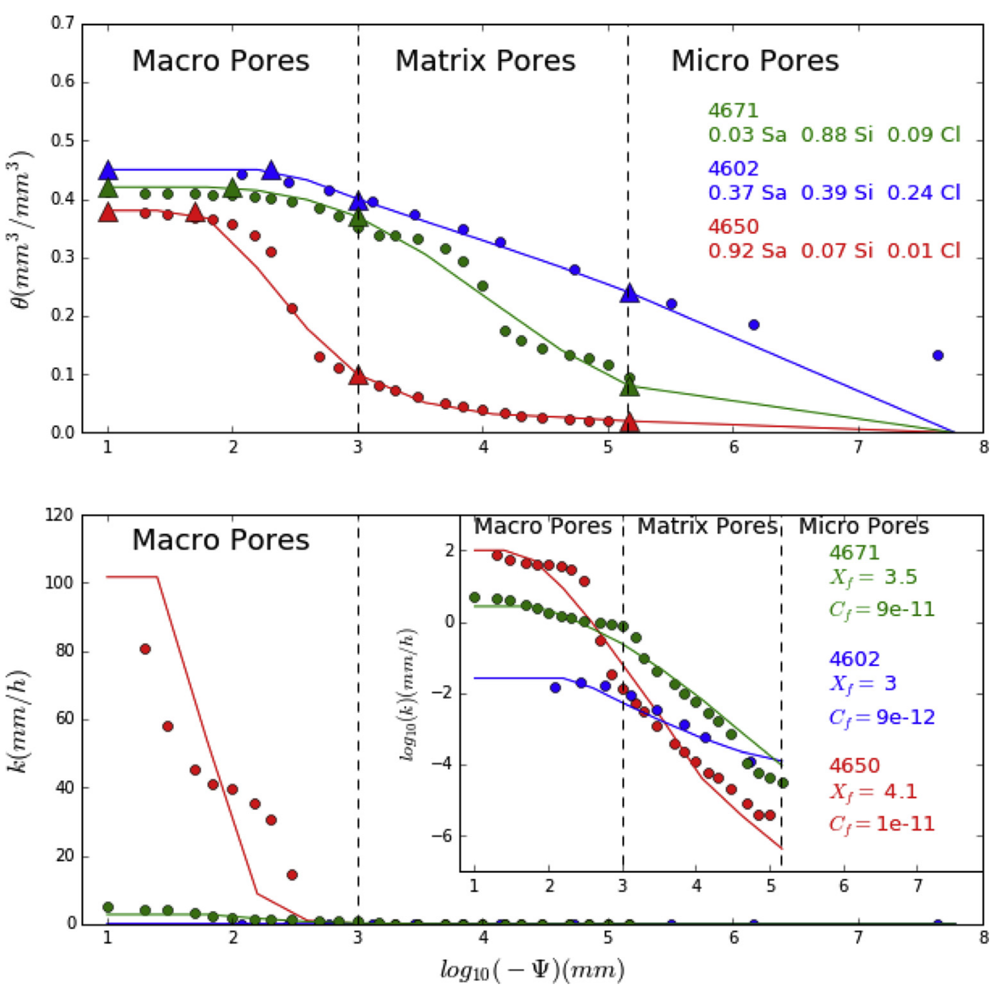

Fig. 2. Top: Examples of the fitted splines fitted using the five-point Hermite spline to the soil-water characteristic $\theta(\log_{10}\psi)$ of three different soils from the UNSODA database. The text in the top panel gives the UNSODA ID for each soil and their corresponding sand, silt and clay proportions. Soil 4671 is a silt, 4602 is a loam and 4650 is a sand. Round markers are values from the UNSODA data-base, and the triangles are the points through which the five-point spline was fitted. The lines are the fitted splines. Bottom: The hydraulic conductivity function $k(\log_{10}\psi)$, and $\log_{10} k(\log_{10}\psi)$ (inset), of the same soils. Text in the top panel gives the UNSODA ID for each soil and their corresponding sand, silt and clay proportions. The text in the bottom panel gives the values of the fitted X_f and C_f for each soil.

would be predicted from the hydrophilic capillary attraction resulting from the soil's pore-size distribution (Doerr et al., 2000). Our approach is simply to assign a reduction factor, R_F, to the sorptivity that is characterised by the relative water content at which water repellency commences, R_{upper}, and R_{lower}, a relative water content at which the greatest repellency occurs, ***min***R_F. Thus R_F ranges from 1 down to ***min***R_F, with a linear decline from R_{upper} to R_{lower} (Fig. 4). Using these three parameters to characterise water repellency is rational, as measurement techniques exist for their estimation (Wijewardana et al., 2015). The value of ***min***R_F can be zero, as even if $R_F = 0$ upon wetting, there will be some, albeit

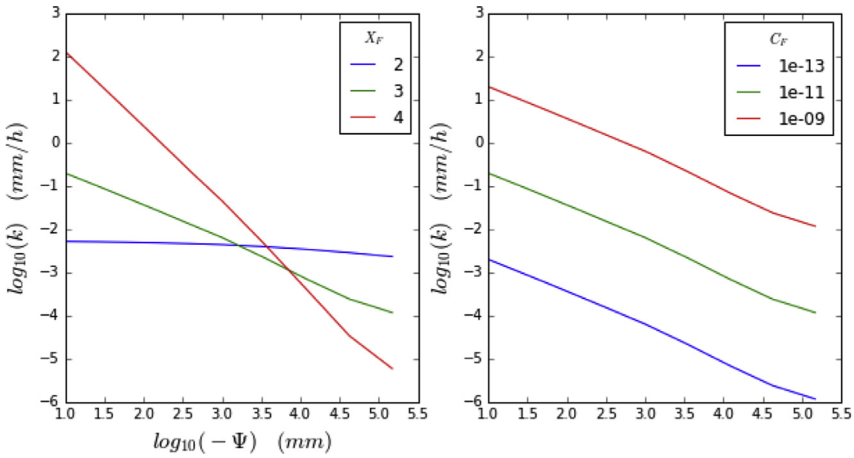

Fig. 3. The hydraulic conductivity function $k(\psi)$ of three hypothetical soils in response to a) a threefold variation in X_F, with $C_F = 1e^{-11}$, and air-entry pressure head $h_{bub} = -10$ mm; and b) a threefold variation in C_F, with $X_F = 3$, and $h_{bub} = -10$ mm.

slow breakdown in water repellency, as Poiseuille flow by gravity will still slowly wet the pores of the matrix.

2.5. The soil-water balance

The flow of water into a soil layer is $k_{in(l)}$ being the sum of the Poiseuille flow by gravity and the capillary attraction by the sorptivity, multiplied by the hydrophobicity factor R_F. The flow-out of the layer $k_{out(l)}$ is due to gravity flow if the pores' diameters are large and surface tension is insufficient. The WEIRDO model calculates the water balance of a layer by summing the change in water content of all the pore classes in hourly time steps.

2.5.1. Infiltration

Infiltration is computed in two steps. First potential infiltration is calculated, then actual infiltration is computed. In this way, surface ponding and a perched water table can be handled if the surface or any underlying layer does not have the capacity to absorb or transmit the water supplied from above.

To calculate potential infiltration, the model steps through each layer, l, and determines the layers capacity to absorb $AbCap_l$ and transmit $TrCap_l$ water:

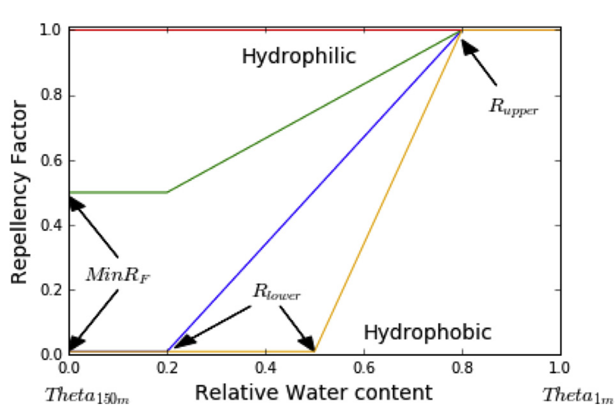

Fig. 4. Examples of trends in the repellency factor, R_F as it affect reduces the sorptivity used to model water flow into hydrophobic pores. The upper red line is a hydrophilic soil, R_F, where the sorption is simply controlled by the porosity. The yellow line is a soil that quickly becomes hydrophobic as the soil dries down. The blue line is a soil that becomes hydrophobic slowly as the soils dries but becomes severely hydrophobic when the soil is dry. The green line represents a soil that does not become severely hydrophobic, but does exhibit sub-critical hydrophobicity as it dries. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

$$AbCap_l = \sum_{c=0}^{9} AbCap_{(l,c)}, \tag{1}$$

where

$$AbCap_{(l,c)} = \min\left(k_{in(l,c)}, D_{A(l,c)}\right), \tag{2}$$

where $k_{in(l,c)}$ is the flow into the pore and $D_{a(l,c)}$ is the depth of air in the pore, representing the pore volume that may accept the infiltrating water

$$TrCap_l = \sum_{c=0}^{9} k_{out(l,c)}, \tag{3}$$

where $k_{out(l,c)}$ is the flow-out of a pore cohort. Then WEIRDO steps through each layer from the bottom layer up to determine how much water may percolate to the layer below $PercCap_l$. For each layer this is calculated as:

$$PercCap_l = AbCap_{l+1} + \min(TrCap_{l+1}, PercCap_{l+1}). \tag{4}$$

The resulting value of $PercCap_l$ for $l = 0$ (the surface layer) represents the potential infiltration into the profile in that time step. Application of Equation (4) poses a problem as the profile has a finite number of layers and starting at the lower-most layer ($l = Nl\text{-}1$), and layer $l + 1$ does not exist! To overcome this a sub-profile conductance is specified to represent the percolation capacity of the soil below the profile.

Rainfall and irrigation are added to the surface pond and the potential infiltration is moved out of the pond and into surface layer. Within each layer, infiltration is partitioned into the smallest pore first. When $AbCap_{(l,c)}$ is full, additional water is then placed sequentially into the larger pores. The $AbCap_{(l,c)}$ of a particular pore is the minimum of the $k_{in(l,c)}$ and $D_{A(l,c)}$. Infiltration events that exceed the $k_{in(l,c)}$ of smaller pores will then bypass them, flowing instead into the larger pores where water is not held as strongly by gravity, and water may flow to deeper layers. This modelling protocol simulates bypass flow when the conductivity into the matrix is less that the size of the infiltration event.

2.5.2. Drainage

Drainage is the next process calculated. Firstly WEIRDO steps through each layer from the top down, and calculates how much water may drain from it:

$$PotDrain_l = \sum_{c=0}^{9} \min\left(k_{out(l,c)}, D_{w(l,c)}\right) \tag{5}$$

The actual drainage from each layer is the minimum of $PotDrain_l$ and $PercCap_{l+1}$. If underlying layers are restricting drainage, then saturated water may perch within the profile. Water draining from each layer is

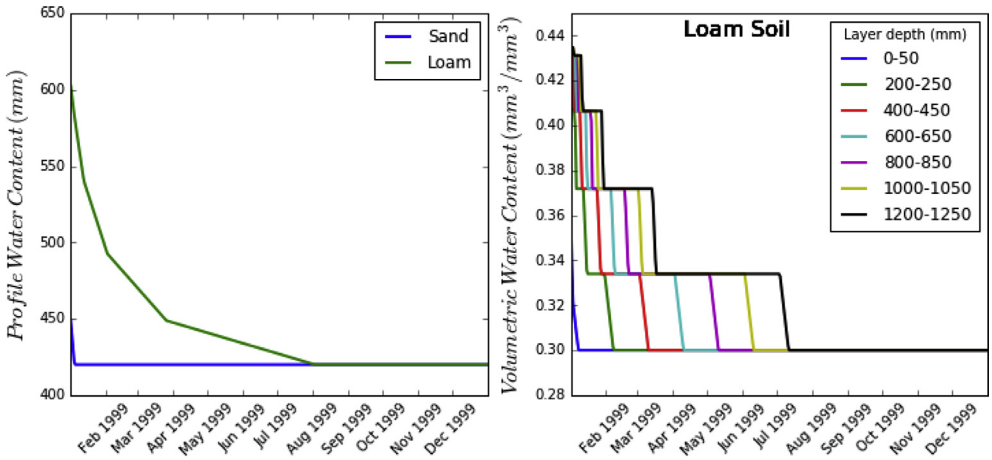

Fig. 5. Left. The profile water content (0–1400 mm) for two soils of different hydraulic properties initialised at saturation and left to drain without further wetting, or evaporation. The silt was not added to the graph because its profile water storage was very similar to the sand. Right. The time course in the draining water content of the different layers for the loam.

distributed through the lower layers using the same method for distributing infiltration.

2.5.3. Redistribution

Redistribution by soil-water diffusion into, or from, each layer is the net result of capillary movement into a layer ($DiffUp_l$) and downward loss out of the same layer ($DiffDown_l$) as follows:.

$$DiffUp_l = \min(PotDiffUp_l, DiffCapUp_l) \tag{6}$$

$$DiffDown_l = \min(PotDiffDown_l, DiffCapDown_l) \tag{7}$$

where

$$PotDiffUp_l = \sum_{c=0}^{9} k_{diff(l-1,c)} \tag{8}$$

is the potential capillary flow into the current layer from the layer below. Then

$$DiffCap_l = \sum_{c=0}^{9} D_{D(l,c)} \tag{9}$$

is the capacity of this layer to receive upward capillarity. Then

$$PotDiffDown_l = \sum_{c=0}^{9} k_{diff(l,c)} \tag{10}$$

is the potential capillary flow out of the current layer to the layer below. And

$$DiffCap_l = \sum_{c=0}^{9} D_{D(l-1,c)} \tag{11}$$

is the capacity of the layer below to receive downward capillary flow. This approach is somewhat analogous to the calculation of the net radiative heat-balance of an object. This approach is not strictly physically correct, as water only diffuses in one direction along a potential gradient. However, calculating redistribution in this way saves the need to determine potential gradients. It produces the same effect of water moving away from wet layers toward dry layers.

2.5.4. Transpiration

The reference transpiration is calculated by APSIM's microclimate module and this is used by WEIRDO to set an upper limit to each hour's transpiration. Then WEIRDO works from largest to smallest pore and the shallowest to deepest layer, removing water at the potential extraction rate from each pore ($WEpot_{(l,c)}$) until the transpiration demand is met. This order is based on the assumption that the plant will first use the water from the largest pores and will also extract water via the path of least resistance which is the shallowest layers (Brown et al., 2009). The term $WEpot_{(l,c)}$ is the product of the water content in a pore cohort, and an uptake proportion (P_U) and a scaling factor to account for large negative water potential in the root (Table 2). The uptake proportion is calculated from $kpoi_{(l,c)}$, divided by the mean distance that water must diffuse to reach an absorbing root ($Dd\mu$). This is a function of root length density (*rld*). Here P_U is analogous to the *kl* parameter used in many tipping-bucket models (Monteith, 1986; Passioura, 1983; Meinke et al., 1993). But here we apply it to individual pores and calculate it as a function of k_{poi} and *rld*. This makes P_U an emergent property of the model that captures the effects of different soil and crop combinations. The cost of this enhancement is that sensible estimations of *rld* must be obtained, something that is not often well addressed in crop models.

2.5.5. Soil-water evaporation

Soil-water evaporation is assumed to occur only from the surface pond, if there is one, and the top layer of the soil profile. Evaporative losses from deeper in the profile are possible as water will redistribute upwards by capillarity into the surface layer as it dries. The potential evaporation rate is calculated using the Penman equation (French and Legg, 1979) which sets an upper limit for evaporation. The radiation and wind terms in the potential rate of soil-water evaporation are multiplied by (1-*Cover*), where *Cover* is the total soil cover from the leaf area and any surface residues. Evaporation is the minimum of the potential rate, the amount of water in the surface pond, and the top layer.

3. WEIRDO within APSIM Next Gen

APSIM simulations using WEIRDO can be set up in the user interface in the same way as for simulations using SoilWat. The main difference is the SoilWater and Water nodes are removed from the Soil, and four other nodes are added:

1. A WEIRDO node which initialises the WEIRDO model and holds its parameters (Tables 1 and 2).
2. A LayerStructure node which determines the number and thickness of the layers that represent the soil.
3. An MRSpline node which fits moisture characteristic curve (Fig. 1) to the parameters provided in the WEIRDO node.
4. An Evapotranspiration node containing a reference ETo model

WEIRDO has been developed to integrate with the APSIM environment to interact with the Manager, Crop and Soil modules. A limitation to the full application of WEIRDO is that solute transport has not yet been

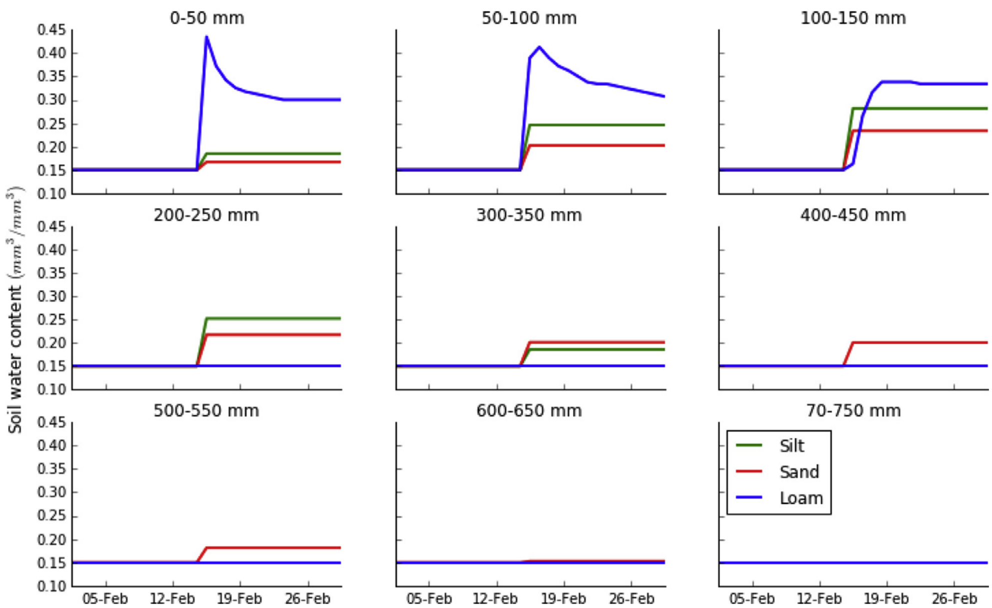

Fig. 6. Infiltration patterns simulated for soils of three different hydraulic properties. Some 30 mm of irrigation was applied to these three soils on 15 February, which would be enough to return the top four layers (0–200 mm) to the field capacity of 0.3 $mm^3\,mm^{-3}$ if there were no preferential flows. The increase in the soil water content in layers deeper than 200 mm for the silt and sand represents the simulation of preferential flow.

coded, so the redistribution of mineral N will not occur.

4. Model application and sensitivities

A series of sensitivity tests shows how WEIRDO simulates flow in structured, hydrophobic soils. These tests were done using three homogeneous, 1.4 m deep soils with the C_F and X_F values of those fitted to the sand, silt and loam soils shown in Fig. 2. Each soil was separated into two 25 mm layers at the surface underlain by 50 mm layers to the bottom of the profile. Values of θ_S, θ_D and θ_L of 0,44, 0.3 and 0.15 $m^3\,m^{-3}$ were used for all three soils and all layers so comparisons show the effects of conductivity parameters alone. Values for h_{bub}, R_{upper}, R_{lower}, and $MinR_F$ were −20 0.8, 0.2 and 1.0, respectively. The processes of infiltration, drainage, evaporation, transpiration and redistribution could be switched off in these tests. This selective switching will not be possible in the released version.

4.1. Drainage

Evaporation, infiltration, transpiration and redistribution processes were all switched off and simulations were initialised with the soils at saturation to assess the drainage patterns. The profiles with sand and silt conductivities (Fig. 5a) quickly drained to a stable water content after 2 days, whereas the loam conductivity took 7 months. The water content of individual layers is shown for the loam where the top layer is the first to drain to stable water content, with successively deeper layers taking longer to drain. The water content patterns are quite blocky. This is because the modelled soil is homogeneous with depth so the water drained from a given pore cohort in a given layer will be replaced by the same amount of water draining from the same pore cohort in the overlying layer until the overlying pore cohort is empty. Following this, the underlying layer's pore cohort will then start to drain.

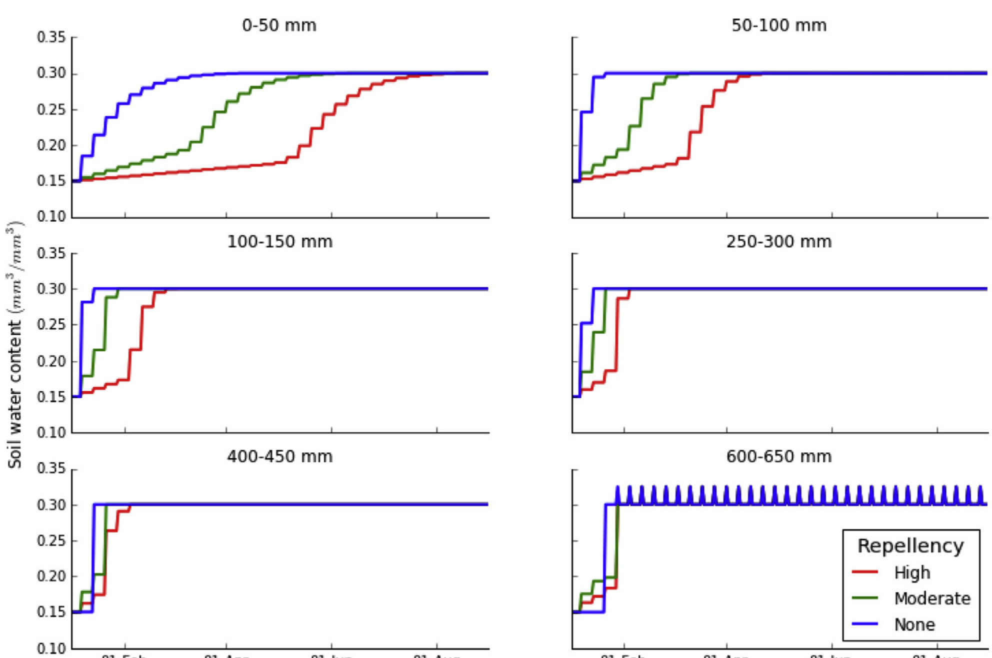

Fig. 7. The temporal dynamics in the rise in the soil water content of a soil exhibiting three different water-repellency characteristics: High, Moderate and None. The soils started at a water content of 0.15 $m^3\,m^{-3}$, and were irrigated weekly with 30 mm of water. The field capacity is 0.3 $m^3\,m^{-3}$ for all soil depths.

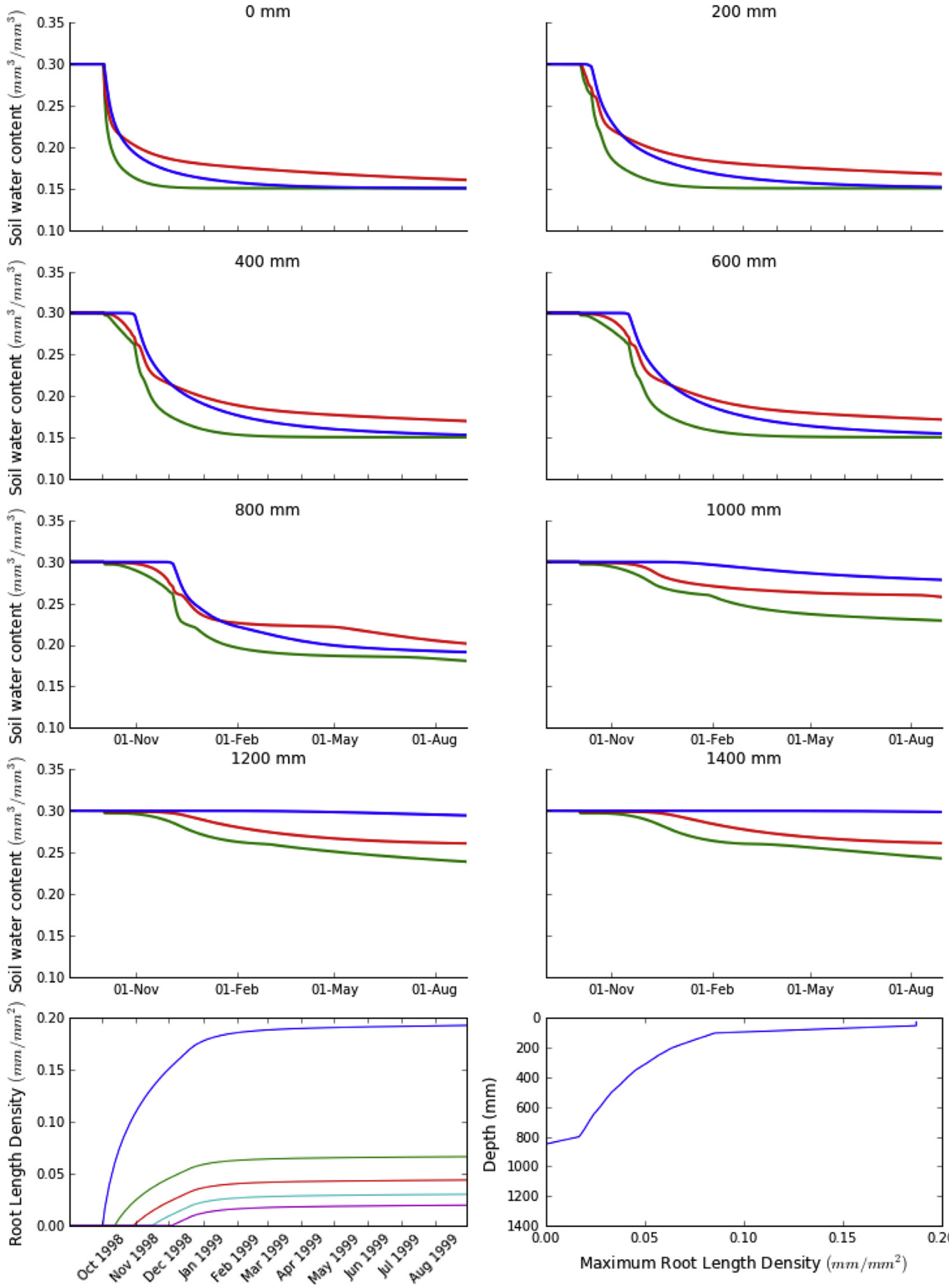

Fig. 8. The soil water content in different layers of sand (red), silt (green) and loam (blue) with a crop planted on 1 September. The bottom-left panel shows the temporal pattern of root length density at the depths of 25, 200, 400, 600 and 800 mm (blue, green, red, cyan and magenta respectively). The bottom-right panel shows the depth distribution of the maximum root length density. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

4.2. *Infiltration*

The primary motivation for the development of WEIRDO was to simulate preferential flow phenomena, where water begins entering a lower layer before the overlying layer has reached field capacity. A simple numerical experiment was conducted where simulations were started with the soil at θ_L. Evaporation, transpiration and redistribution processes were all switched off. An irrigation of 30 mm was applied, enough to return the top 200 mm to θ_D if all the irrigation water were all retained in the top four layers. Irrigation was applied at a rate of 60 mm h^{-1}. Simulations were run using the three different soils. For the loam, with its low values of X_f, the simulated infiltration pattern is the same as expected from a tipping-bucket model, with all the water retained in the top 200 mm (Fig. 6). However, the silt and the sand did not retain all the irrigation in the top layers with some water infiltrating to the 200–400 mm layer for the silt, and some infiltrating into to the

400–600 mm layer for the sand. This is the preferential flow phenomenon we set out to model and which was characterised in the field under sprinkler irrigation by Clothier and Heiler (1983).

4.3. Repellency

For our hydrophobicity test, we considered three degrees of water repellency: High, Moderate and No repellency, where $MinR_F$ was 0.01, 0.05 and 1.0 respectively. These changes were imposed on a soil with an X_F of 3.5, which is one with an intermediate propensity for water to be absorbed into the matrix and not lead to bypass flow. The soil was initialised at the soil water content, θ_L, in all layers and irrigations of 30 mm were applied every 7 days at a rate of 60 mm h^{-1}. Evaporation and redistribution processes were switched off to highlight the impact of water repellency.

The No repellency simulation quickly rewet the top layer, with wetting progressing to deeper layers once the top layer reached θ_D (Fig. 7). Here, the wetting front did not reach the deepest layer (600–650 cm) until the sixth irrigation event, and the drainage started after the eighth wetting event. For the Moderate repellency simulation, the water content rose more slowly in the top layer, and the wetting front penetrated deeper into the soil sooner. With the High repellency simulation, the rise in the surface soil-water content was slow, and a considerable amount of water drained from the bottom of this profile, even before the upper layers had fully wetted. For the surface layer of this severely repellent soil, R_{upper}, the fully hydrophilic condition, was not achieved until August. This simulated behaviour mimics the temporal dynamics that have been observed for hydrophobic pastoral soils (Müller et al., 2014).

4.4. Transpiration

To test the transpiration routine, a set of simulations was run again using the three different soil types. The SLURP model (Teixeira et al., 2017) within APSIM was used to represent pasture with a maximum rooting depth of 80 cm, and a root front extension rate of 15 mm d^{-1}, a final LAI of 5.0, and a final height of 0.4 m. All processes were switched on, apart from soil-water evaporation so that the patterns simulated were due to water extraction alone. Water extraction began in the top layer immediately following sowing on 1 October (Fig. 8). Extraction was fastest for the silt and the sand soils. However, the conductivity in the matrix drops fastest for the sand and soon its water extraction rates declined. The drying front moved down the profile faster than the roots in the sand and silt, but not for the loam. Robertson et al. (1993) also observed drying fronts moving down the soil faster than the root front. Although the roots did not extend below 80 cm, small amounts of water extraction were modelled in deeper layers as water redistributed up into the drying layers above.

5. Model testing and validation

We tested our WEIRDO model against soil-water content data measured under a field trial with alfalfa that was irrigated using three different schedules: no irrigation, irrigation replacing evapotranspiration losses twice a week, and irrigation returning the soil to 'refill point' every three weeks. The trial has been described in detail by Michel et al. (2014). The soil-water content was measured in 20 cm depth increments using CS616 Campbell reflectometer probes in the top 20 cm, and a CPN neutron probe in lower layers down to a depth of 1.5 m, and recorded every 7–14 days during the trial. The field experiment covered three seasons from 2011 through until 2014. The crop cover and root-water uptake dynamics were simulated in APSIM using the simple plant model, SLURP (Teixeira et al., 2017). Here we compare the field measurements of the soil water content with the predictions by WEIRDO with all processes switched on. The soil was considered to be hydrophilic.

The comparison between the soil-water content measurements made by Michel et al. (2014) and the predictions using WEIRDO are shown in Fig. 9. The total profile water storage is well predicted by WEIRDO (Fig. 9, top left), as are the predictions by WEIRDO of the water content in the soil layers.

6. Conclusions

In modelling the soil-water dynamics of Earth's critical zone, there are two major challenges. The first is to understand and model the processes of infiltration, runoff and redistribution in structured soils that exhibit preferential flow through macropores. The other challenge is to

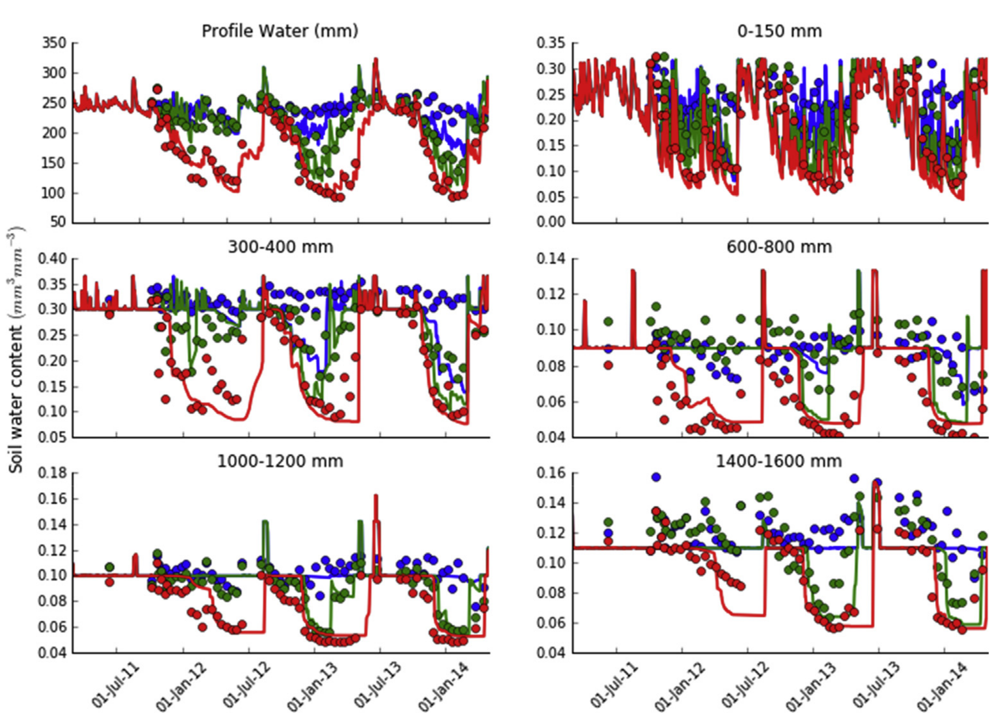

Fig. 9. The profile soil water content (top left panel), and the volumetric water content of different layers throughout the profile measured under frequent (blue), infrequent (green) and no (red) irrigation treatments from the alfalfa field trial of Teixeira et al. (2017). The circles are the measured values and lines are the simulations by WEIRDO. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

parametrise and model the impact of hydrophobicity. Here we have developed a new modelling approach, which is based on physical principles yet having functionality to enable easier parameterisation to predict soil-water dynamics in structured soils displaying ephemeral hydrophobicity. Our model, WEIRDO (**W**ater **E**vaporation **I**nfiltration **R**edistribution **D**rainage run**O**ff), has been developed in the APSIM Next Generation platform. We have carried sensitivity tests to show how WEIRDO predicts infiltration, drainage, redistribution, transpiration and soil-water evaporation for three distinctly different soil textures displaying differing hydraulic properties. Finally, we have validated WEIRDO by comparing its predictions against three years of soil-water content measurements made under an irrigated alfalfa trial. These results provide validation of the models ability to simulate soil-water dynamics in structured soils.

Acknowledgements

This research was carried out under the “Maximising the Value of Irrigation” (CO9X1309) programme with principal funding from the Ministry of Business, Innovation and Employment and co-funding from the Foundation for Arable Research, Horticulture New Zealand, Environment Canterbury, Hawke's Bay Regional Council and Irrigation New Zealand. The programme is led by Landcare Research, PFR and FAR. We are grateful for the insightful comments provided by Dr Carolyn Hedley and Dr Joseph Pollacco of Landcare Research.

Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.cageo.2018.01.014.

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